Week-11

1. In the context of data privacy, which right requires companies to delete user data when requested?
    a. Right to intellectual property retention
    b. Right to be forgotten
    c. Right to data portability
    d. Right to transparency

    Explanation: AI Governance [9:56-11:50] - The right to be forgotten (or the right to erasure) gives the data subject the right to erasure of their personal data.

2. Which OECD AI Principle states that AI systems should operate safely and function properly under normal use, misuse, or adverse conditions?
    a. Inclusive growth, sustainable development and well-being
    b. Transparency and explainability
    c. Robustness, security, and safety
    d. Accountability

    Explanation: AI Governance [18:12-19:00] - Robustness, Security, and Safety states that AI systems should be robust, secure, and safe throughout their entire lifecycle.

3. Under the EU AI Act, which application category is classified as an 'unacceptable risk'? **(Select all that apply.)**
    a. Social Scoring Systems
    b. Critical Infrastructure Management
    c. Law Enforcement
    d. Cognitive Behavioural Manipulation

    Explanation: AI Governance [25:22-26:56] - According to the EU AI Act, social scoring and cognitive behavioural manipulation are unacceptable risks.

4. When a system like an email spam server filters out malicious messages in the background without requiring user intervention, which type of threat mitigation is it demonstrating?
    a. Educating and training users
    b. Providing interface-level threat warnings
    c. Silently eliminating the threat
    d. Enforcing legal regulations

    Explanation: AI Governance [4:41-8:17] - Spam servers operate behind the scenes to remove unwanted or dangerous emails automatically.

5. What specific transparency requirement does the EU AI Act impose on deepfakes or generated media?
    a. They must be exclusively hosted on European-based servers.
    b. They must be clearly labeled as AI-generated.
    c. They must not be published on the internet at all.
    d. They must be heavily taxed before distribution.

Explanation: AI Governance [28:00-29:35] - The act mandates that any images, audio, or video files generated or modified by AI must be explicitly labeled.

6. In the AI Value Chain, who is considered the "deployer"?
    a. The person or organization that uses the AI systems under its own authority.
    b. The original engineering team that built the AI model.
    c. The individual who imports the AI system into the EU market.
    d. The government official responsible for writing the regulations.

    Explanation: AI Governance [30:12-31:55]: The deployer is the persons that use AI systems under their authority.

7. To satisfy the OECD principle of "accountability," AI actors should ensure "traceability" throughout the system's lifecycle. What does this specifically help achieve?
    a. It allows companies to track user information that can lead to their location, such as IP addresses.
    b. It ensures one can trace back how data was collected and how it influences decision-making.
    c. It guarantees the system uses minimal computing power and has a low carbon footprint.
    d. It allows the AI to completely wipe its training period after a set period.

    Explanation: AI Governance [19:02-19:53]: Traceability ensures that if an AI makes a particular choice, auditors can trace the datasets and processes that led to that specific outcome.

8. In the SaGE framework, how is semantic inconsistency functionally defined?
    a. Failing to identify commonsense reasoning violations within a single prompt.
    b. Generating factually incorrect statements in response to a factual prompt.
    c. Utilizing grammatically disparate structures to answer identical prompts.
    d. Providing semantically disparate answers to semantically equivalent prompts.

    Explanation: Research Overview: SaGE- Quantifying moral consistency in LLMs [2:27-2:52] - Semantic inconsistency is an LLM generating contradictory or different answers when presented with semantically equivalent prompts.

9. In the SaGE framework, what does a higher semantic graph entropy indicate about the LLM?
    a. The model is exhibiting higher semantic consistency.
    b. The model is exhibiting higher semantic inconsistency.
    c. The model is exhibiting lower syntactic divergence.
    d. The model is exhibiting lower fundamental judgment.

    Explanation: Research Overview: SaGE- Quantifying moral consistency in LLMs [15:32-18:43]: Higher graph entropy indicates greater randomness among the generated responses, which means the model's answers are inconsistent.

10. A developer attempts to reduce an LLM's tendency to give contradictory answers to moral questions by lowering the generation temperature. Based on SaGE, what is the expected outcome?
    a. The consistency score will improve to near perfection.
    b. The consistency score will perfectly match human scores.
    c. The consistency score will drop to near zero.
    d. The consistency score will remain largely unaffected.

    Explanation: Research Overview: SaGE- Quantifying moral consistency in LLMs [21:52-22:44]: The SaGE study demonstrated that changing the temperature does not significantly alter the model's consistency.

11. According to the SaGE framework, how does an LLM's accuracy on standard commonsense reasoning benchmarks correlate with its moral consistency?
    a. Accuracy negatively correlates with consistency.
    b. Accuracy does not correlate with consistency.
    c. Accuracy strictly correlates with consistency.
    d. Accuracy conditionally correlates with consistency.

    Explanation: Research Overview: SaGE- Quantifying moral consistency in LLMs [22:45-23:58]: The SaGE study demonstrated that a model's benchmark accuracy does not correlate with its consistency at all.